

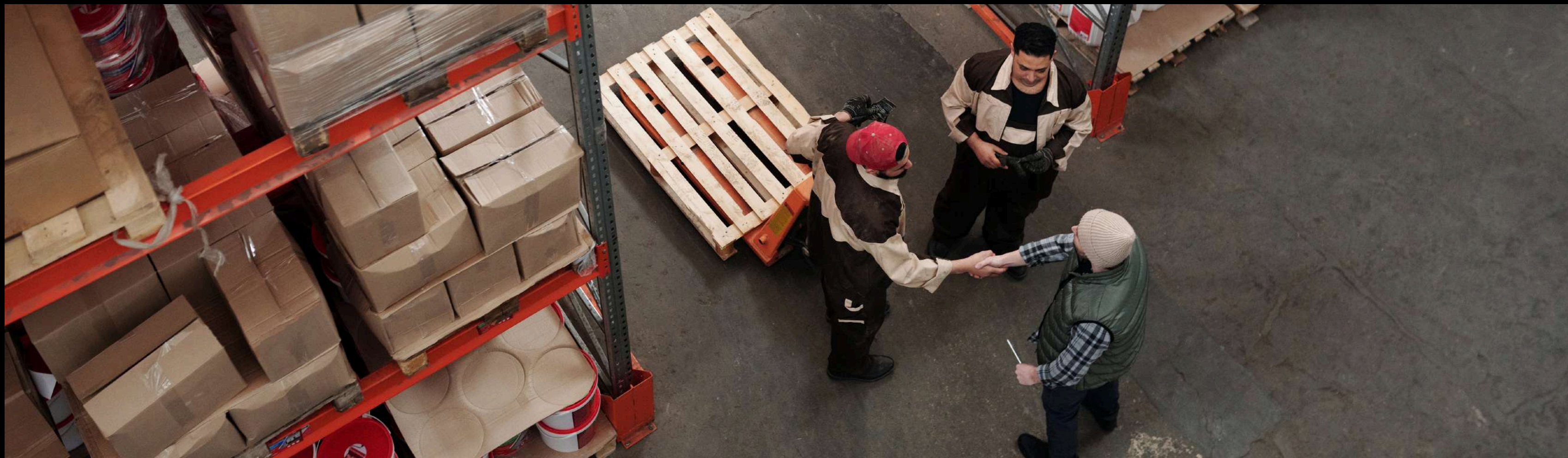
IBM BOB HACKATHON



PORTPULSE

CONTAINER CONGESTION PREDICTOR &
PORT OPERATIONS OPTIMISER

*An Overview of the
solution prepared by
PC Pirates*



PORTS ARE FLYING BLIND UNTIL IT'S TOO LATE

THE PROBLEM



- 2021 LA/Long Beach backlog: 100+ ships waiting offshore for weeks
- Global economic impact: \$10B+ in supply chain losses
- Root cause: berths, cranes, and yard space are still allocated manually in spreadsheets
- Congestion is spotted reactively — after vessels are already queuing
- Rerouting decisions arrive too late to matter



OUR SOLUTION

→ INTRODUCING PORTPULSE

- A PREDICTIVE, ROLE-BASED PORT OPERATIONS PLATFORM THAT TURNS REACTIVE FIREFIGHTING INTO PROACTIVE PLANNING
- LIVE AT: PORTPULSE
- COMBINES FORECASTING, PRESCRIPTIVE RECOMMENDATIONS, AND REAL-TIME OPERATIONAL TOOLS IN ONE DASHBOARD
- BUILT FOR THE PEOPLE WHO ACTUALLY RUN THE PORT FLOOR — NOT JUST ANALYSTS



THE ASK

THE CHALLENGE DEMANDS

- PREDICT CONGESTION HOTSPOTS FROM VESSEL SCHEDULES + BERTH CAPACITY DATA
- RECOMMEND ALTERNATE ROUTING STRATEGIES
- OPTIMISE BERTH AND CRANE ASSIGNMENTS
- GENERATE A 72-HOUR PORT OPERATIONS PLAN FOR SHIFT SUPERVISORS

ROLE-BASED ACCESS

ONE PLATFORM, EVERY STAKEHOLDER

- *ADMINISTRATOR — MANAGES USERS, MASTER DATA, AND SYSTEM CONFIGURATION*
- *SHIFT SUPERVISOR — EXECUTES THE 72-HOUR OPERATIONS PLAN*
- *VESSEL PLANNER — MANAGES SCHEDULES AND ROUTING DECISIONS*
- *TERMINAL MANAGER — OVERSEES BERTH, CRANE, AND YARD OPERATIONS*
- *EACH ROLE SEES A TAILORED VIEW — RIGHT INFORMATION, RIGHT PERSON, RIGHT TIME*



CONGESTION PREDICTION & 72-HOUR OPERATIONS PLAN

SEEING THE BACKLOG BEFORE IT FORMS

- CONGESTION HEATMAP — VISUALISES HOTSPOT RISK ACROSS BERTHS AND YARD ZONES IN REAL TIME
- PREDICTIVE MODEL FLAGS HIGH-RISK WINDOWS USING VESSEL SCHEDULES AND BERTH CAPACITY DATA
- SHIFTS DECISION-MAKING FROM REACTIVE TO ANTICIPATORY

FROM PREDICTION TO ACTION

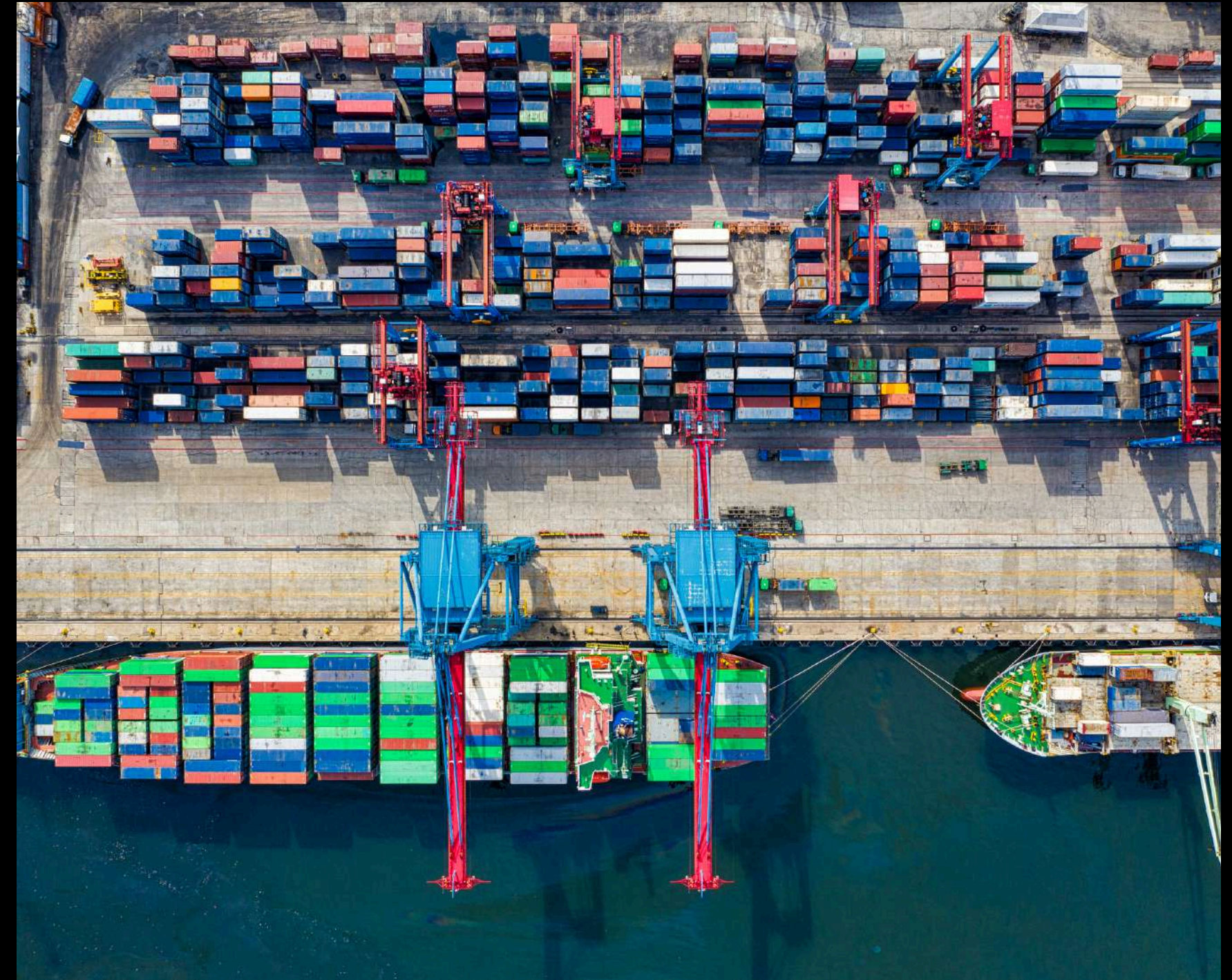
- AUTO-GENERATED 72-HOUR PLAN TAILORED FOR SHIFT SUPERVISORS
- TRANSLATES CONGESTION FORECASTS INTO A CONCRETE OPERATIONAL SCHEDULE
- REMOVES THE MANUAL SPREADSHEET PLANNING BOTTLENECK AT THE CORE OF THE PROBLEM STATEMENT



BERTH & RESOURCE OPTIMISATION

OPTIMISING THE PHYSICAL OPERATION

- *BERTH ALLOCATOR — INTELLIGENTLY ASSIGNS BERTHS ACROSS VESSELS TO MINIMISE CONGESTION*
- *TERMINAL MAP — VISUAL, SPATIAL VIEW OF TERMINAL LAYOUT AND VESSEL/YARD STATUS*
- *LIVE QUEUE — REAL-TIME TRACKING OF VESSELS WAITING OR IN PROCESS*
- *TOGETHER, THESE REPLACE MANUAL SPREADSHEET-BASED BERTH/CRANE ALLOCATION*



NOT JUST ALERTS — RECOMMENDATIONS

- *SYSTEM DOESN'T STOP AT FLAGGING RISK — IT PRESCRIBES SPECIFIC CORRECTIVE ACTIONS*
- *SUGGESTS ALTERNATE ROUTING STRATEGIES TO AVOID PREDICTED BOTTLENECKS*
- *CONVERTS INSIGHT INTO AN EXECUTABLE NEXT STEP FOR OPERATORS*

PRESCRIPTIVE ACTIONS



SIMULATION & DECISION SUPPORT

TESTING DECISIONS BEFORE COMMITTING

- *DELAY SIMULATOR — MODELS THE DOWNSTREAM IMPACT OF VESSEL OR BERTH DELAYS BEFORE THEY HAPPEN*
- *LETS PLANNERS STRESS-TEST "WHAT-IF" SCENARIOS AHEAD OF REAL DISRUPTION*
- *ASK AI — CONVERSATIONAL ASSISTANT FOR QUICK, CONTEXTUAL OPERATIONAL QUERIES*



MONITORING & ANALYTICS

KEEPING OPERATIONS ACCOUNTABLE

- *ACTIVITY LOG — FULL AUDIT TRAIL OF ACTIONS AND CHANGES ACROSS THE PLATFORM*
- *BENCHMARKS — PERFORMANCE TRACKING AGAINST OPERATIONAL TARGETS/HISTORICAL BASELINES*
- *AUTO SYNC — KEEPS VESSEL, BERTH, AND SCHEDULE DATA CONTINUOUSLY UP TO DATE*

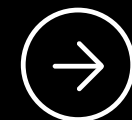
USABILITY & DATA MANAGEMENT

- *GUIDED TOUR — IN-APP WALKTHROUGH THAT DOUBLES AS A LIVE USER MANUAL*
- *DARK MODE / LIGHT MODE — ACCESSIBLE FOR ROUND-THE-CLOCK PORT OPERATIONS*
- *MASTER DATA MANAGEMENT — CENTRALISED CONTROL OF CORE REFERENCE DATA*
- *ADMIN CAN CREATE AND MANAGE NEW USERS*
- *EXPORT TO CSV AND PRINT SUPPORT — FOR REPORTING, COMPLIANCE, AND OFFLINE USE*



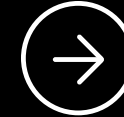
IMPACT

- CONVERTS WEEKS OF REACTIVE CONGESTION RESPONSE INTO HOURS OF PROACTIVE PLANNING
- DIRECTLY TARGETS THE MANUAL, SPREADSHEET-DRIVEN ALLOCATION PROCESS THAT CAUSED THE 2021 BACKLOG
- SCALABLE ACROSS PORTS OF ANY SIZE — SMALL TERMINALS TO MAJOR HUBS LIKE LA/LONG BEACH
- REDUCES VESSEL WAIT TIME, DEMURRAGE COSTS, AND DOWNSTREAM SUPPLY CHAIN DISRUPTION



FRONTEND

- CORE LIBRARY: REACT 18
- LANGUAGE: TYPESCRIPT
- BUILD TOOL: VITE
- STYLING: TAILWIND CSS
- MAPPING UI: LEAFLET AND REACT-LEAFLET (FOR INTERACTIVE MAPS)
- ICONS: LUCIDE REACT



BACKEND

- FRAMEWORK: FASTAPI (PYTHON) RUNNING ON UVICORN
- DATABASE ORM: SQLALCHEMY
- DATABASE: POSTGRESQL (INDICATED BY PSYCOPG2-BINARY) ALONG WITH SQLITE FOR LOCAL DEVELOPMENT (PORTPULSE.DB)
- DATA VALIDATION & SETTINGS: PYDANTIC
- AUTHENTICATION: JWT & PASSWORD HASHING USING PYTHON-JOSE AND PASSLIB
- DATA SCIENCE / MACHINE LEARNING: NUMPY, SCIPY, AND SCIKIT-LEARN (LIKELY USED FOR THE CONGESTION PREDICTION AND OPTIMIZATION LOGIC)
- TESTING: PYTEST AND PYTEST-ASYNCIO

TECH HIGHLIGHTS



THANK YOU



LIVE DEMO: [CLICK HERE](#)

THE PEOPLE BEHIND IT

- Naitik Patel
- Shah Diya Rajan
- Brij Patel
- Dodiya Rudra Ketansinh

